CHEMISTRY TARGET : JEE- Advanced	CAPS-18 METALLURGY D-AND F-BLOCK ELEMENTS

SINGLE CHOICE QUESTIONS

1. When MnO_2 is fused with KOH in the presence of air, the product formed is :

(A) purple colour $KMnO_4$ (B) green cololur K_2MnO_4

(C) colourless MnO_4^- (D) purple colour K_2MnO_4

2. In which of the following species, the colour is due to charge transfer.

(I) $[Mn(OH)_4]^{2-}$ (II) MnO_4^{2-}

(III) MnO_2 (IV) $KMnO_4$

(A) I, II, III correct (B) II, IV correct (C) I, III correct (D) only IV correct

3. XCl_2 (excess) + $YCl_2 \rightarrow XCl_4 + Y$;

$YO \xrightarrow[>400^\circ]{\Delta} \frac{1}{2}O_2 + Y$, Ore of Y would be:

(A) Siderite (B) Cinnabar (C) Malachite (D) Hornsilver

4. Choose the correct code regarding Roasting process.

(I) It is the process of heating ore in air to obtain the oxide

(II) It is an exothermic process

(III) It may be used for hydrated oxide and oxysalt ore to remove impurities

(IV) It is used after the concentration of ore

(A) I, II and III (B) I, II and IV

(C) I, III and IV (D) I, II, III and IV

5. Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false.

(i) In Goldschmidt thermite process aluminium acts as a reducing agent.

(ii) Mg is extracted by electrolysis of aq. solution of MgCl2

(iii) Extraction of Pb is possible by carbon reduction method

(iv) Red Bauxite is purified by Serpeck's process

(A) TTTF (B) TFFT (C) FTTT (D) TFTF

6. Which of the following statement is/are incorrect?

(A) Hall and Heroult's method is used in extraction of Al.

(B) Iron obtained from blast furnace is cast iron.

(C) Wrought iron is obtained by puddling process.

(D) None of these

7. $(T)\xrightarrow{\text{compound(U)+conc. } H_2SO_4}(V)$Red gas

$\xrightarrow{NaOH+AgNO_3}(W)$Red ppt.$\xrightarrow{NH_3\text{soln.}}(X)$

T is an Orange crystalline compound which when burnt imparts violet color.

(W)Red ppt.$\xrightarrow{\text{dil. HCl}}(Y)$white ppt.

$(U)\xrightarrow{NaOH\Delta}(Z)$gas (gives white fumes with HCl) sublimes on heating

Identify (T) to (Z).

(A) T = $KMnO_4$, U = HCl, V = Cl_2, W = HgI_2, X = $Hg(NH_2)NO_3$, Y = Hg_2Cl_2, Z = N_2

(B) T= $K_2Cr_2O_7$, U = NH_4Cl, V = CrO_2Cl_2, W= Ag_2CrO_4, X = $[Ag(NH_3)_2]+$, Y= AgCl, Z = NH_3

(C) T = K_2CrO_4, U = KCl, V = CrO_2Cl_2, W = HgI_2, X = Na_2CrO_4, Y = $BaCO_3$, Z = NH_4Cl

(D) T = K_2MnO_4, U = NaCl, V = CrO_3, W = $AgNO_2$, X = $(NH_4)_2CrO_4$, Y = $CaCO_3$, Z = SO_2

8. Metal-metal bonding is more frequent in 4d or 5d series than in 3d series due to:

(A) their greater enthalpies of atomisation

(B) the large size of the orbitals which participates in the metal-metal bond formation

(C) their ability to involve both ns and (n – 1) d electrons in the bond formation

(D) the comparable size of 4d and 5d series elements

MCQ (One or more than one correct) :

9. Select correct statement regarding silver extraction process.

(A) When the lead-silver alloy is rich in silver, lead is removed by the cupellation process

(B) When the lead-silver alloy is rich in lead, lead is removed by Parke's or pattinson' s process

(C) Zinc forms an alloy with lead, from which lead is separated by distillation

(D) Zinc forms an alloy with silver, from which zinc is separated by distillation

10. Correct statements about forth floatation process used in metallurgy

(A) It is based on the difference in wettability of different minerals

(B) Sodium ethyl xanthate is used as collector

(C) NaCN is used as a depressant in the mixture of ZnS and PbS when ZnS forms soluble complex

(D) This method is generally employed to dress the sulphide ores

11. Which of the following is true for calcination of a metal ore ?

(A) It makes the ore more porous.

(B) The ore is heated to a temperature when fusion just begins.

(C) Hydrated salts lose their water of crystallisation.

(D) Impurities of S, As and Sb are removed in the form of their volatile oxides.

12. Which of the following statements regarding d-block elements are true?

(A) the colour of anhydrous $CuSO_4$ is blue

(B) "splitting of silver" can be prevented by covering the surface of molten silver with charcoal

(C) Iodine liberated in a reaction can be estimated by titration against a standard thiosulphate solution

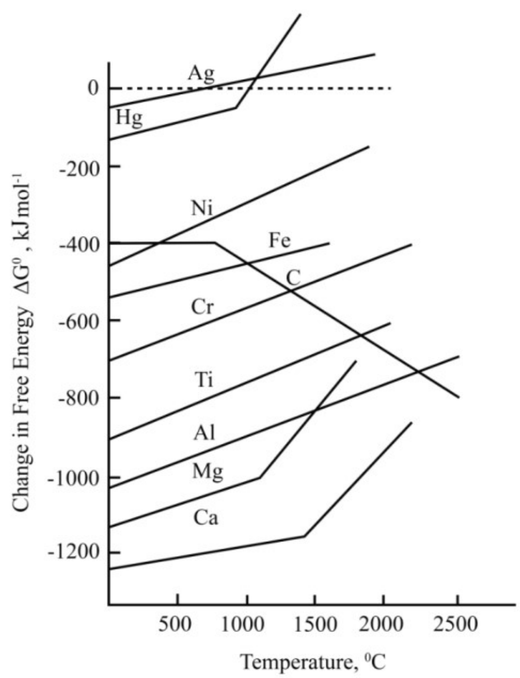
(D) Lanthanum is first element of third transition series.

Comprehension Type Question:

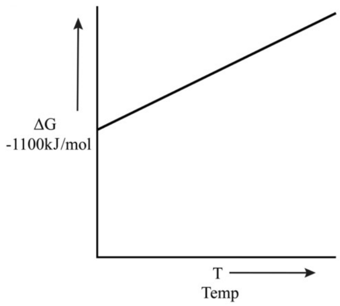
Comprehension-1

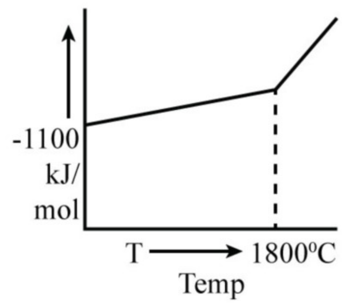
For a spontaneous reaction, the free energy change must be negative. $\Delta G = \Delta H - T\Delta S$, ΔH is the enthalpy change during the reaction. T is the absolute temperature, and ΔS is the change in entropy during the reaction. Consider a reaction such as the formation of an oxide $M + O_2 \rightarrow MO$ Dioxygen is used up in the course of this reaction. Gases have a more random structure (less ordered) than liquid or solids. Consequently, gases have a higher entropy than liquids and solids. In this reaction S (entropy or randomness) decreases, hence ΔS is negative. Thus, if the temperature is raised then $T\Delta S$ becomes more negative. Since, $T\Delta$ is subtracted in the equation, then ΔG becomes less negative. Thus, the free energy change increases with the increase in temperature.

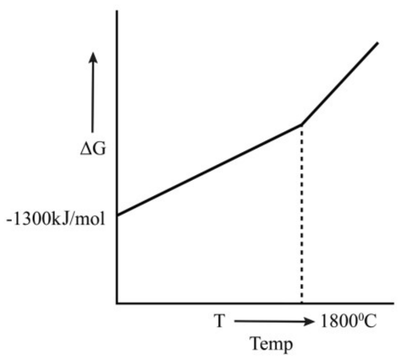
The free energy changes that occur when one mole of common reactant (in this case dioxygen) is used may be plotted graphically against temperature for a number of reactions of metals to their oxides. The following plot is called an Ellingham diagram for metal oxide. Understanding of Ellingham diagram is extremely important for the efficient extraction of metals.

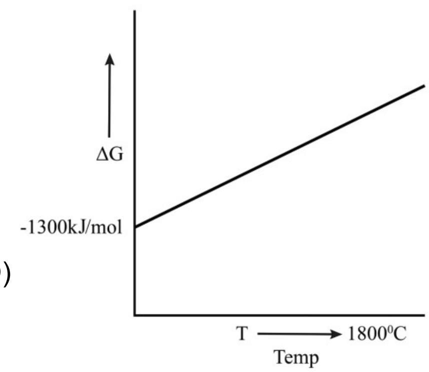


13. For the conversion of Ca(s) to CaO(s) which of the following represent the ΔG vs T?

(A)


(B)


(C)


(D)


14. Free energy change of Hg and Mg for the conversion to oxides the slope of ΔG vs. T has been changed above the boiling points of the given metal because

(A) above the boiling point of the metal, entropy is increased

(B) above the boiling point of the metal, entropy is decreased

(C) above the boiling point of the metal, entropy change is equal to zero

(D) All of these

15. Which of the following elements can be prepared by heating the oxide above 400°C?

(A) Hg (B) Mg (C) Fe (D) Al

16. As per the Ellingham diagram of oxides which of the following conclusion is true?

(A) Al reduces Fe_2O_3, whereas MgO cannot be reduced by Al at 1500ºC

(B) Fe reduces Al_2O_3, whereas MgO cannot be reduced by Al at 1500ºC

(C) Al reduces Fe_2O_3, whereas MgO cannot be reduced by Ca at 1500ºC

(D) Al can reduce both Fe_2O_3 and MgO to the corresponding metal at 1500ºC

Comprehension-2

Magnesium is a valuable, light weight metal used as a structural material as well as in alloys, batteries, and in chemical synthesis. Although magnesium is plentiful in Earth's crust, it is mainly found in the sea water (after sodium). There is about 1.3 g of magnesium in every kilogram of sea water. The process for obtaining magnesium from sea water employs all three types of reactions, i.e., precipitation, acid-base, and redox reactions.

17. Precipitation reaction involves formation of

(A) insoluble $MgCO_3$ by adding Na_2CO_3

(B) insoluble $Mg(OH)_2$ by adding $Ca(OH)_2$

(C) insoluble in $MgSO_4$ by adding Na_2SO_4

(D) insoluble $MgCl_2$ by adding NaCl

18. Acid-base reaction involves reaction between

(A) $MgCO_3$ and HCl

(B) $Mg(OH)_2$ and H_2SO_4

(C) $Mg(OH)_2$ and HCl

(D) $MgCO_3$ and H_2SO_4

19. Redox reaction takes place

(in the extraction of Mg)

(A) in the electrolytic cell when fused $MgCl_2$ is subjected to electrolysis

(B) when fused $MgCO_3$ is heated

(C) when fused $MgCO_3$ is strongly heated

(D) none of the above

INTEGER

20. How many of the following metals can be extracted by carbon reduction method?

Na, Al, Mg, Sn, Fe, Pb, Zn, Ca

21. No. of following reagents in which ppt A is soluble is :

$AgNO_3 + NaCl \longrightarrow A$

(i) aq NH_3 (ii) aq. KCN (iii) $Na_2S_2O_3$

(iv) Conc. HCl (v) aq. NaOH (iv) aq. KI

22. Column-I and Column-II contains four entries each. Entries of column-I are to be match with some entries of column-II. Each entry of column-I may have the matching with one or more than one entries of column-II.

Column-I	Column-II
(A) Hydrometallurgy applied in commercial extraction of metal	(p) Ag
(B) carbon reduction applied in commercial extraction in metal	(q) Zn
(C) Aqueous salt solution is used in electrolytic Refining method	(r) Sn
(D) Metal present in anode mud of refining of crude copper	(s) Au
	(t) Cu

(A) A $\rightarrow$ P, S; B $\rightarrow$ Q; C $\rightarrow$ P, Q; D $\rightarrow$ P, S
(B) A $\rightarrow$ P, S; B $\rightarrow$ Q, R; C $\rightarrow$ P, S; D $\rightarrow$ P, Q, R, S, T
(C) A $\rightarrow$ Q, R; B $\rightarrow$ P, S; C $\rightarrow$ P, Q, R, S, T; D $\rightarrow$ P, S
(D) A $\rightarrow$ P, S, T; B $\rightarrow$ Q, R; C $\rightarrow$ P, Q, R, S, T; D $\rightarrow$ P, S

23. Column-I contains four statements following reason and Column-II consists of four options P, Q, R, S Answer the following :

P $\rightarrow$ If both statement and reason are true and reason is correct explanation of statement.

Q $\rightarrow$ If both statement and reason are true and reason is not correct explanation of statement.

R $\rightarrow$ If statement is correct and reason is incorrect.

S $\rightarrow$ If both statement and reason are incorrect.

Column-I	Column-II
(A) **Statement :** The reaction of oxalic acid with acidified $KMnO_4$ is first slow and then speeds up by itself. **Reason :** $KMnO_4$ decomposes into MnO_2 in sunlight.	(P)
(B) **Statement :** Anh, $ZnCl_2$ can't be made by heating $ZnCl_2 .2H_2O$. **Reason :** It undergoes hydrolysis to produce Zn(OH)2 and HCl.	(Q)
(C) **Statement :** $KMnO_4$ is not used as a primary standard substance **Reason :** It is deliquescent in nature.	(R)
(D) **Statement :** $K_2Cr_2O_7$ has orange colour due to polarisation. **Reason :** In dichromate ion all Cr—O bonds are identical.	(S)

24. A white substance (A) reacts with dilute H_2SO_4 to produce a colourless suffocating gas (B) and a colourless solution (C). The reaction of gas (B) with potassium iodate and starch solution produces a blue colour solution. Aqueous solution of (A) gives a white precipitate with $BaCl_2$ solution which is soluble in dilute HCl. Addition of aqueous NH_3 or NaOH to (C) produces first a precipitate which dissolves in excess of the respective reagent to produce a clear solution. Similarly, addition of excess of potassium ferrocyanide to (C) produces a precipitate (D) which also dissolves in aqueous NaOH giving a clear solution. Identify (A), (B), (C) and (D). Write the equations of the reactions involved.

25. Explain the large difference in melting point of Cr(1920ºC) and Zn(420ºC).

ANSWER KEY CAPS-18

1.	(B)	**2.**	(B)	**3.**	(B)	**4.**	(D)	**5.**	(D)
6.	(B)	**7.**	(B)	**8.**	(A)	**9.**	(ABD)	**10.**	(ABCD)
11.	(AC)	**12.**	(BCD)	**13.**	(B)	**14.**	(A)	**15.**	(A)
16.	(A)	**17.**	(B)	**18.**	(C)	**19.**	(A)	**20.**	(4)
21.	4 (i) (ii) (iii) (iv)			**22.**	(D)				
23.	(A) - (q) ; (B) - (p) ; (C) - (r) ; (D) - (s)								